Subject-Disjoint Benchmarking of Nano Detectors for Visible Driver-Cue Localization

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***Abstract*—Object detection can localize visible evidence inside a vehicle cabin, but detection accuracy alone does not establish a temporal driver state. We benchmark single-frame localization of four discrete, momentary driver cues under a fixed subject-disjoint protocol. The primary red–green–blue (RGB) experiment retains all 15,723 sampled Driver Monitoring Dataset (DMD) frames from 14 drivers, including 12,722 negatives, and evaluates You Only Look Once (YOLO) 11n, YOLO26n, and the nano Fine-grained Distribution Refinement (D-FINE-N) detector across three predeclared seeds. Confidence selection uses validation data only. Protected testing reports Common Objects in Context (COCO) mean average precision (mAP), micro- and macro-F_1, false detections on negative frames, and synchronized batch-1 latency. For the completed YOLO runs, YOLO11n reaches $91.79 \pm 0.44\%$ mAP_{50}, $83.42 \pm 1.36\%$ macro-F_1, and 68.0 ± 1.3 frames per second (FPS). YOLO26n instead reaches the higher $\mathrm{mAP}_{50:95}$ of $56.98 \pm 0.81\%$. These directions hold for every paired seed, while subject-level F_1 reveals substantially larger variation than seed-level standard deviations. D-FINE-N results and a separate exploratory near-infrared (NIR) training-negative exposure study are [PLACEHOLDER]. The benchmark therefore supports a bounded comparison of complete detector systems for visible cue localization, not temporal fatigue or behavior inference.**

***Index Terms*—driver monitoring, object detection, subject-disjoint evaluation, visible driver cues, near-infrared imaging, real-time inference**

I. Introduction

Driver monitoring systems (DMSs) operate under changes in driver identity, illumination, viewpoint, and occlusion. Whole-frame recognition can describe an activity, but it may also exploit identity or cabin context without showing the visual evidence used for a decision. Object detection provides a complementary formulation: it localizes cue-specific evidence, makes false positives and misses spatially interpretable, and measures detections on frames containing no target cue. A single-frame detector also avoids the latency and architectural complexity of a temporal model when real-time throughput is the experimental focus. This formulation does *not* infer fatigue, intent, duration, or another temporal driver state.

DMD provides multimodal attention and alertness recordings [1], while Drive&Act contains synchronized color, NIR, depth, and pose streams for fine-grained activities [2]. Both contain temporally correlated video and repeated observations of each driver. Evaluation can therefore be optimistic if neighboring frames or identities cross partitions, natural negative frames are removed, the test set determines the confidence threshold, or latency boundaries differ.

Recent work proposes specialized lightweight networks for in-cabin recognition and detection [3]–[6]. Our objective is different: to compare complete, publicly released nano detector systems without task-specific architectural modification. The contributions are:

- a primary subject-disjoint RGB benchmark for four explicitly defined visible cues, with all sampled frames preserved and annotation structure automatically audited;
- validation-only operating-point selection followed by protected testing, three paired RGB seeds, class- and subject-level analysis, false detections on negative frames, and synchronized latency; and
- a checksum-backed reproduction workflow plus a clearly separated, one-seed NIR training-negative exposure study with deterministic 1-FPS midpoint sampling.

II. Related Work

A. Driver Recognition and Cue Localization

MobileVGG and convolution–transformer hybrids demonstrate efficient distracted-driver classification [3], [7]. PO-GUISE+ uses pose and object guidance for efficient driver-action recognition in Drive&Act NIR imagery [8]. Such whole-frame or temporal recognition tasks are not interchangeable with the present bounding-box task. Here, a correct prediction must identify both a cue class and its spatial evidence in one frame; no temporal state is estimated.

Detection-specific work modifies YOLO backbones, attention, or multiscale aggregation for dangerous-driving or fatigue cues [4]–[6]. Comparative studies assess convolutional neural networks (CNNs) against transformers [9] or several YOLO generations [10]. However, differences in driver separation, retained negatives, operating-point calibration, and runtime protocol limit direct cross-paper ranking.

B. Released Detector Systems

The Detection Transformer (DETR) casts object detection as bipartite set prediction [11]; the Real-Time Detection Transformer (RT-DETR) makes this formulation practical for real-time use [12]; and D-FINE refines box-edge distributions for improved localization [13]. We compare D-FINE-N with YOLO11n and YOLO26n as shipped systems, including their

TABLE I
FROZEN RGB BOX SEMANTICS AND TEST SUPPORT

Cue	Bounding-box target	Test n	Median area
Yawning	Mouth	33	1.38%
Hand over mouth	Full face and covering hand	28	24.67%
Drinking	Visible hand/container–mouth interaction	56	16.09%
Phone use	Visible hand–phone interaction	497	8.37%

native training and inference recipes. The pinned YOLO26 implementation uses end-to-end one-to-one predictions without non-maximum suppression (NMS), whereas YOLO11n uses its conventional NMS path [14]. This is therefore a system-level benchmark, not an isolation of architecture alone.

III. METHODOLOGY

A. Task Definition and Workflow

The task is bounding-box localization of four visible RGB cues: yawning, hand over mouth, drinking, and phone use. Detection is used because it spatially localizes the evidence, discourages whole-frame identity and cabin shortcuts, and exposes errors on negative frames. Figure 1 summarizes the shared protocol. Dataset construction, splitting, training, and validation precede a checksum and confirmation gate; no test-derived threshold or model change is permitted. The RGB benchmark is primary. NIR is a distinct secondary experiment rather than a paired modality comparison.

B. RGB Dataset and Annotation Protocol

The RGB dataset contains 15,723 face crops sampled at 1 FPS from 81 DMD recordings of 14 drivers. One primary annotator labeled the data in a single smooth pass; no independent second-person review was established. Every sampled frame was preserved to avoid post-sampling curation, yielding 3,001 positive and 12,722 negative frames. Table I freezes the spatial ontology. Box semantics intentionally differ because the visible evidence differs by cue.

Each frame contains zero or one box. Yawning and hand-over-mouth are mutually exclusive; when a hand covers a yawn, hand-over-mouth takes precedence. Boxes enclose only the visible target. Occluded, truncated, or out-of-frame anatomy is never estimated, although a partially occluded cue remains valid when visually identifiable. No arbitrary minimum-pixel cutoff is applied at 640×640.

An automated COCO audit found 15,723 unique images and 3,001 unique annotations, with zero duplicate identifiers or file names, orphan annotations, invalid categories, nonfinite or nonpositive boxes, out-of-bounds boxes, or multi-annotation frames. The smallest annotated area was 0.81% of the frame; 139 of 159 yawning boxes occupied below 2%. This is a structural audit, not evidence of inter-annotator agreement or semantic correctness.

C. Split, Models, and Training

An exhaustive 8:3:3 subject assignment minimizes the worst relative class-distribution deviation, with root-mean-square deviation as a tie-breaker. The frozen training, validation, and test partitions contain 9,087, 3,423, and 3,213 frames. The three test subjects contribute 1,080, 920, and 1,213 frames and 178, 183, and 253 positives, respectively.

COCO-pretrained YOLO11n, YOLO26n [14], and D-FINE-N [13] use 640×640 input, physical batch size 8, four-step gradient accumulation for an effective batch of 32, 220 epochs, automatic mixed precision (AMP), and no early stopping. A fixed patch normalizes an incomplete final accumulation window by its actual sample count. Native recipes are retained: Ultralytics `optimizer=auto` with a linear schedule, and the official D-FINE AdamW/MultiStepLR recipe. Seeds $\{13, 37, 73\}$ were declared in advance, and no run was selected for presentation.

D. Validation, Test Metrics, and Profiling

The best model-native validation checkpoint is retained. Its confidence threshold τ^* maximizes validation micro-F_1 over $[0.01, 0.99]$, with higher precision and then higher τ as tie-breakers. At an intersection over union (IoU) of 0.5, predictions are greedily matched one-to-one within class. Protected testing reports COCO mAP_{50} and $\mathrm{mAP}_{50:95}$ [15], precision, recall, micro- and macro-F_1, and

$$\mathrm{FD}_{100} = 100 \frac{\mathrm{FP}_{\mathrm{neg}}}{N_{\mathrm{neg}}}, \tag{1}$$

where $\mathrm{FP}_{\mathrm{neg}}$ is the number of above-threshold false-positive detections on negative frames. Thus, FD_{100} is a static per-frame count, not a modeled nuisance-alert rate; no temporal confirmation or debouncing is applied.

RGB results are the sample mean and sample standard deviation (SD) over three seeds. Because the subject split is fixed, SD measures optimization variability, not generalization uncertainty across held-out drivers. Each test subject is therefore reported separately, and paired per-seed differences are checked for sign consistency. No significance test is claimed for $n = 3$.

Batch-1 16-bit floating-point (FP16) profiling uses an NVIDIA GeForce RTX 4060 8-GB graphics processing unit (GPU). CUDA synchronization brackets model-forward and tensor-to-final-detection intervals. Disk input/output, decoding, preprocessing, and metric computation are excluded. We report p50 tensor-to-final latency and sustained FPS; these measurements do not represent central-processing-unit or embedded-device deployment.

E. Exploratory NIR Protocol

The secondary NIR source pool contains 3,786 one-second front-top active-NIR (850 nm) Drive&Act snippets from 15 drivers. Twenty-three unilluminated task snippets are excluded before splitting, leaving 3,763. Drinking and phone-use tracklets provide two positive classes. Each snippet contributes one deterministic frame at 1 FPS: the temporal midpoint at $t = 0.5$ s, or zero-based source-frame offset 14. The corresponding box is linearly interpolated at the same instant, then the driver crop is resized to 640×640. Unlike RGB, NIR therefore has dataset-specific frame exclusion and interpolated annotations.

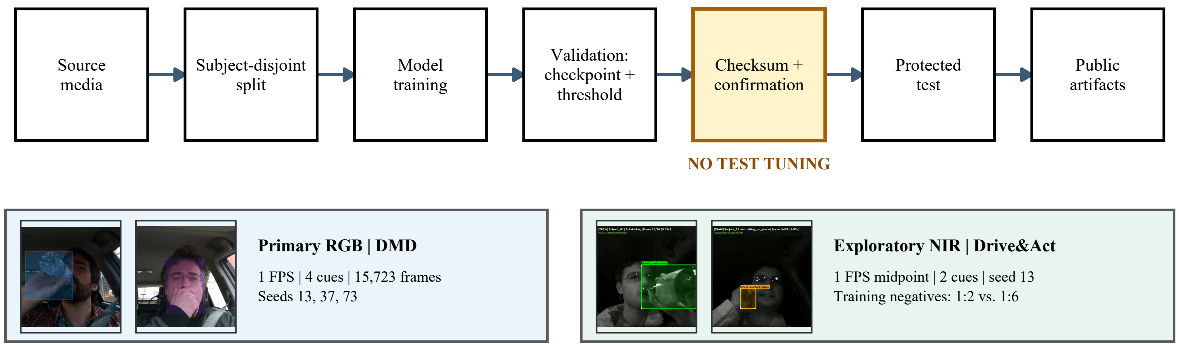


Fig. 1. End-to-end benchmark workflow with representative annotated DMD and Drive&Act frames. RGB and NIR share the validation-before-test gate but retain dataset-specific sampling, ontology, seeds, and training conditions. FPS denotes frames per second. The diagram and all subsequent plots were generated with R/ggplot2.

A fixed 9:3:3 subject partition is shared across all six model–condition runs. Validation and test contain identical ordered examples. Training always contains 270 positive snippets and either 540 or 1,620 negative snippets, corresponding to nominal positive:negative ratios of 1:2 and 1:6. The 1:2 negatives are a subject-stratified seed-13 subset of the 1:6 pool. With epochs fixed, 1:6 has three times the unique negatives and approximately three times the optimization steps. It is consequently a *training-negative exposure study*, not a causal class-ratio ablation under matched training signal.

All NIR jobs use seed 13, and operating metrics follow the RGB protocol. Since one frame represents each snippet, frame- and snippet-level decisions coincide; equivalent snippet metrics are not duplicated in the publication tables.

IV. Results and Analysis

A. Aggregate RGB Results

Table II summarizes the six completed RGB YOLO runs. YOLO11n has higher mAP_{50}, micro-F_1, macro-F_1, and throughput at every paired seed; its p50 latency is 4.67–5.17 ms lower. YOLO26n has higher $\mathrm{mAP}_{50:95}$ at every seed, with paired differences of 0.94–2.43 percentage points, despite lower mAP_{50}. The evidence supports a system-level trade-off between coarse-IoU operating accuracy and stricter box localization, not universal superiority.

B. Class Imbalance and Paired-Seed Evidence

Phone use accounts for 497 of 614 positive test instances, whereas hand over mouth, yawning, and drinking have only 28, 33, and 56 instances. This imbalance can influence learned representations, validation-selected thresholds, and the pooled micro-F_1. It does not directly dominate COCO mAP, because AP is first computed by category and then averaged equally across the four categories.

The class-wise mean differences, YOLO26n minus YOLO11n, are $+6.21$ points for hand over mouth, -3.85 for yawning, $+1.02$ for drinking, and $+3.02$ for phone use. The aggregate $\mathrm{mAP}_{50:95}$ advantage is $+1.60$ points. Across paired seeds, YOLO11n is consistently higher for yawning, while YOLO26n is consistently higher for hand over mouth and phone use. The drinking direction changes with seed, so no consistent difference is claimed for that class. Reporting macro-F_1 and per-class AP beside micro-F_1 prevents the majority phone-use class from silently defining the conclusion, although the wide SD for low-support cues still warrants caution.

C. Held-Out Subject Sensitivity

Averaged across seeds, YOLO11n micro-F_1 is 94.94%, 62.41%, and 94.72% for Subjects 05, 10, and 12; YOLO26n obtains 96.11%, 54.11%, and 92.29%. The between-subject change is far larger than the pooled seed SD. Reporting only the pooled test set would therefore hide the difficult driver.

D. Validation Operating Point

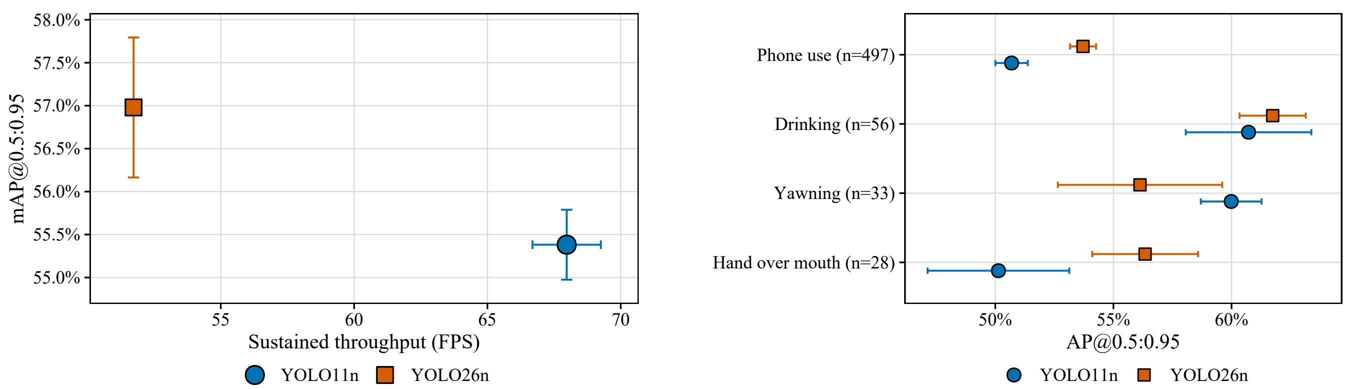
Figure 3 makes the validation-only operating-point decision explicit. Lines and ribbons show the seed mean and sample SD across the full confidence sweep. Each model has one filled marker at its mean validation-selected operating point. YOLO11n selects lower thresholds and gives higher validation micro- and macro-F_1, whereas YOLO26n maintains high F_1 farther into the threshold range. The false-detection panel shows the corresponding reduction on negative validation frames. None of these curves uses the protected test set.

E. Qualitative Error Analysis

Figure 4 shows predeclared seed-13 examples from frozen predictions. Correct detections localize the intended interaction; a shared miss occurs under glare with a small drinking target. More importantly, one nominal negative appears visually cue-like. It may reflect a difficult ontology boundary or a missed label, so the structural annotation audit alone cannot establish semantic validity. The example is retained because it exposes the benchmark's principal annotation risk.

TABLE II
RGB SUBJECT-DISJOINT TEST RESULTS (MEAN ± SAMPLE SD OVER SEEDS 13, 37, AND 73). LATENCY IS p50 TENSOR-TO-FINAL DETECTION; BOLD MARKS THE NUMERICAL BEST PER COLUMN, NOT STATISTICAL SIGNIFICANCE.

Model	mAP_{50}	$\mathrm{mAP}_{50:95}$	Micro-F_1	Macro-F_1	FD_{100}	p50 (ms)	FPS	Params (M)	GFLOPs
YOLO11n	**91.79±0.44**	55.38±0.41	**86.60±0.49**	**83.42±1.36**	**0.63±0.04**	**14.30±0.20**	**68.0±1.3**	2.59	6.44
YOLO26n	89.28±0.99	**56.98±0.81**	84.44±1.14	79.39±1.49	0.67±0.06	19.21±0.09	51.7±0.2	**2.51**	**5.78**
D-FINE-N					**[PLACEHOLDER]**				



(a) Accuracy–throughput trade-off.

(b) Per-class $\mathrm{AP}_{50:95}$ with test support.

Fig. 2. Completed RGB YOLO results. Error bars show sample SD across seeds 13, 37, and 73. Model identities are encoded in a separate legend so labels do not obscure observations. D-FINE-N is added automatically when its aggregate is frozen.

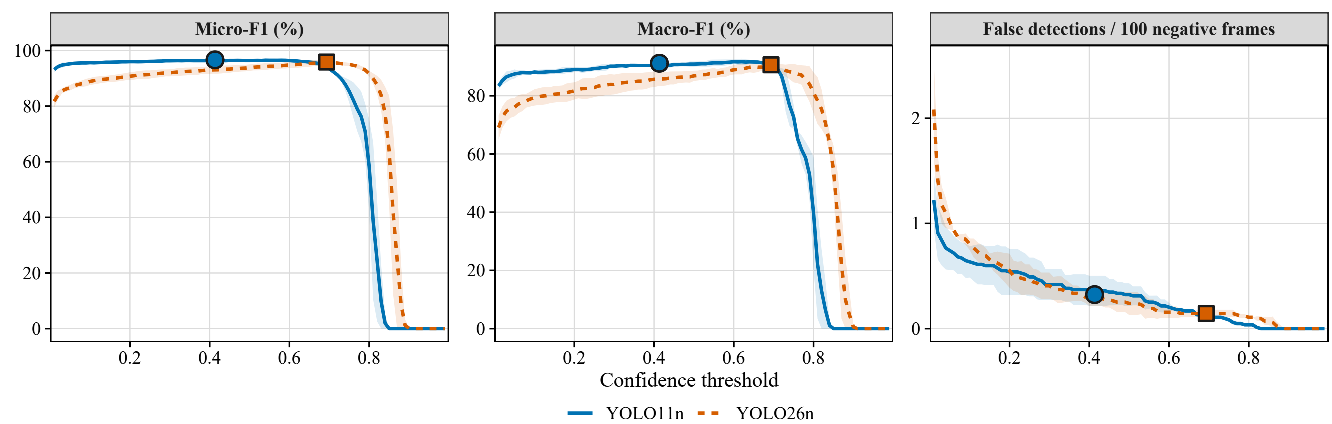


Fig. 3. RGB validation confidence sweep. Lines show the seed mean and translucent ribbons show sample SD. Dark-edged markers identify the mean micro-F_1-selected operating points.

F. Exploratory NIR Results

Table III and Fig. 5 reserve the final six protected-test results. Both artifacts are deliberately marked pending; no value is imputed from RGB or validation. The final analysis will compare each model only between its 1:2 and 1:6 training-negative exposures using the same deterministic midpoint frame for every condition.

TABLE III
EXPLORATORY NIR TEST RESULTS AT SEED 13

Model	Neg.	$\mathrm{mAP}_{50:95}$	Micro-F_1	Macro-F_1	FD_{100}	p50
YOLO11n	1:2		**[PLACEHOLDER]**			
YOLO11n	1:6		**[PLACEHOLDER]**			
YOLO26n	1:2		**[PLACEHOLDER]**			
YOLO26n	1:6		**[PLACEHOLDER]**			
D-FINE-N	1:2		**[PLACEHOLDER]**			
D-FINE-N	1:6		**[PLACEHOLDER]**			

V. DISCUSSION

A. Accuracy–Efficiency Trade-offs

The completed evidence does not identify one universally best YOLO system. YOLO11n is the better operating-point choice when coarse-IoU accuracy, macro-F_1, lower latency, and throughput are prioritized. YOLO26n gives the higher

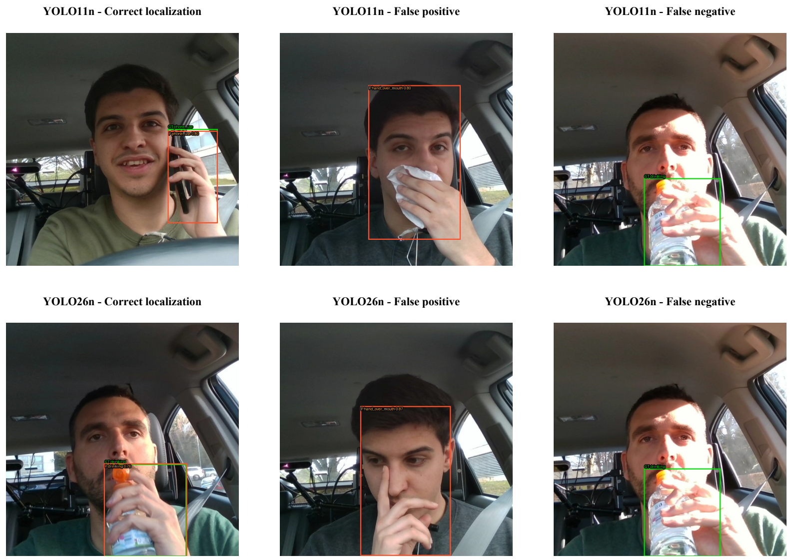


Fig. 4. Representative seed-13 RGB outcomes. Green boxes denote ground truth and orange boxes denote predictions at the validation-selected threshold. Rows are completed detector systems; columns show a true positive, a false detection on a nominal negative, and a false negative.

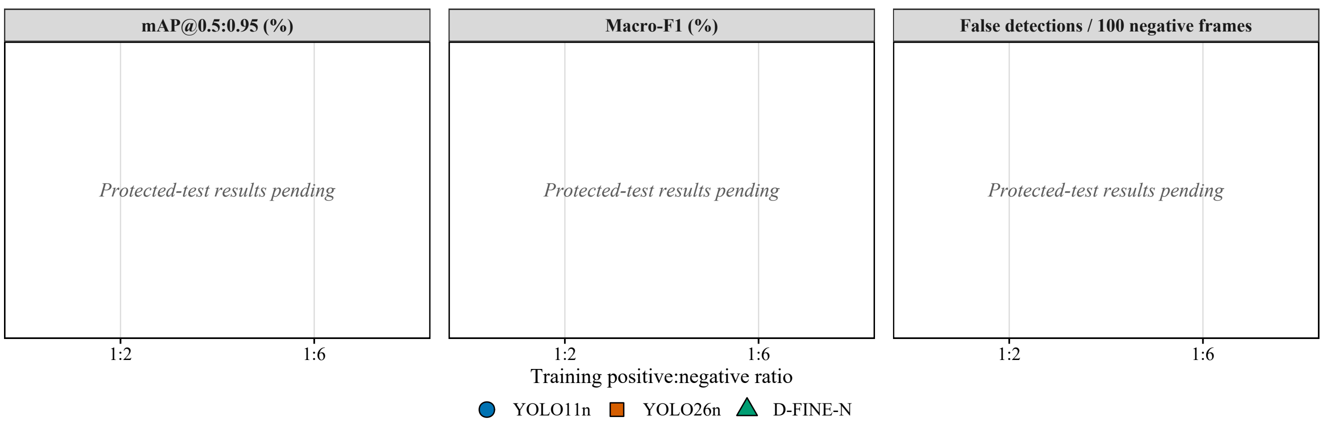


Fig. 5. Final inventory for the exploratory NIR training-negative exposure comparison at deterministic 1-FPS midpoint sampling. The checked-in template remains visibly pending until all six seed-13 protected tests are complete; it will then show strict-IoU accuracy, macro-F_1, and false detections per 100 negative frames.

strict-IoU mAP with slightly fewer parameters and fewer giga floating-point operations (GFLOPs), but its native end-to-end inference path is slower in the measured tensor-to-final boundary. The comparison is useful precisely as a shipped-system benchmark: training and postprocessing remain native, so differences cannot be attributed to architecture alone.

B. Class and Subject Effects

The phone-use majority is a real benchmark characteristic and a limitation for pooled operating metrics, but it does not invalidate YOLO26n's strict-IoU advantage. Equal-weight class AP, macro-F_1, per-class supports, and paired-seed directions separate that result from majority-class dominance. The narrow aggregate margin, low support for three categories, and drinking sign reversal mean the accurate conclusion is a trade-off rather than a definitive rank. The difficult Subject 10 further shows that seed variation on a single split understates deployment variation across people.

C. Interpretation Scope

Subject-disjoint evaluation reduces identity leakage but does not establish population-level generalization from only 14 drivers and three held-out test identities. A single annotator and one pass provide internally consistent data but no independent semantic agreement estimate; the qualitative nominal-negative case makes this concrete. Correlated frames remain within RGB recordings, and single-frame detection neither estimates temporal fatigue nor models alert debouncing. NIR avoids within-snippet duplication through one midpoint frame, but 1-FPS sampling does not characterize motion within the second. NIR additionally differs in dataset, ontology, annotation source, and preprocessing, uses one seed, and changes negative diversity together with training steps. It cannot be interpreted as a controlled RGB-versus-NIR modality effect. Finally, RTX 4060 timing excludes preprocessing and is not an embedded-hardware measurement. These boundaries define what the benchmark supports without removing its value as a transparent, reproducible comparison.

VI. Conclusion and Future Work

The completed RGB YOLO comparison establishes a reproducible system-level trade-off: YOLO11n provides the stronger validation-calibrated operating point and throughput, while YOLO26n provides higher strict-IoU AP. Paired seeds support those aggregate directions, but class and subject analyses prevent an overgeneralized winner claim. The final three-model RGB conclusion after D-FINE-N and the separate NIR exposure findings are **[PLACEHOLDER]**.

Future work will add independent multi-annotator semantic review, increase rare-cue and held-out-driver coverage, repeat the benchmark across multiple subject partitions and external cabins, and test class-balanced or matched-step training without changing the protected baseline. Temporal confirmation and snippet-level alert logic will connect frame localization to user-facing warnings. The NIR study will be strengthened with matched optimization exposure and additional seeds, while embedded-device profiling will include preprocessing and end-to-end input latency. These extensions directly target the uncertainty exposed by the present class, subject, and qualitative analyses.

Data and code: Authored annotations, frozen partitions and checksums, sanitized predictions, evaluation tools, and publication artifacts are available at https://github.com/IamOumarIbrahim/lightweight-driver-behavior-detection; licensed DMD and Drive&Act media and trained checkpoints remain local under their original access terms.

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